

US Correctional Facilities Master Dataset

Comprehensive Technical Documentation & Methodology Report

EXECUTIVE SUMMARY & KEY DATASET METRICS

- Total Master Facilities: 6,768 unique institutions nationwide
- Geographic Coverage: 55 States & Territories (All 50 states + DC, PR, GU, VI, MP)
- GPS Coordinates Completeness: 100.0% (6,768 / 6,768 facilities validated in WGS84)
- Physical Street Addresses: 100.0% (6,765 verified street addresses)
- Direct Contact Phone Numbers: 6,229 facilities
- Total Reported Design Bed Capacity: 2,411,708 beds
- Total Reported Inmate Population: 2,069,547 inmates

1. Research Context & Data Provenance

The United States correctional infrastructure is decentralized across federal, state, county, municipal, tribal, and private jurisdictions. No single governmental agency maintains a real-time, unified master registry of all facilities. To build this comprehensive national database, data was acquired and synthesized from two authoritative federal sources:

1. Homeland Infrastructure Foundation-Level Data (HIFLD) – Prison Facilities Layer (DHS / FEMA / Oak Ridge National Laboratory):

Provides critical infrastructure geospatial baseline data for secure detention facilities across all 50 states, DC, and territories.

2. Federal Bureau of Prisons (BOP) Official Public Institution Directory (U.S. Department of Justice):

Provides authoritative live operational records for all Federal Correctional Complexes (FCC), US Penitentiaries (USP), Federal Correctional Institutions (FCI), Federal Prison Camps (FPC), Detention Centers (FDC/MDC), and Reentry Offices (RRM).

2. Jurisdictional Breakdown & Summary

The dataset captures 6,768 facilities across six jurisdictional authorities:

Jurisdiction	Facility Count	Design Bed	Reported	Primary Facility
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		Capacity	Population	Classifications
County / Local	3,924	777,361	608,074	County Jails, Adult Detention Centers, Juvenile Facilities
State	2,347	1,364,812	1,223,733	State Prisons, Correctional Institutions, Re-entry Centers
Federal	253	186,134	165,862	Federal Bureau of Prisons (USP, FCI, FPC, FDC, MDC, FMC, RRM)
Municipal / Local	182	38,490	32,841	City Jails, Municipal Holding Facilities
Multi-Jurisdiction	35	16,985	15,221	Regional Jail Authorities, Joint County Compacts
Private / Contract	27	27,926	23,816	Contracted Detention Facilities
Total	6,768	2,411,708	2,069,547	Nationwide Master Total

3. Ingestion, Cleaning & Entity Resolution Methodology

A. Ingestion & API Pagination

Data was extracted from the DHS HIFLD ArcGIS FeatureServer using programmatic HTTP pagination with parameters resultOffset and resultRecordCount=2000 in WGS84 spatial reference (outSR=4326). This bypassed server transfer limits and ensured 100% feature capture. Raw responses were cached deterministically to guarantee reproducibility.

B. Deduplication of Spatial Multi-Part Features

ArcGIS layers frequently represent multi-building complexes or split campus parcels as separate polygon geometries, resulting in duplicate attribute entries. The raw HIFLD layer contained

10,738 entries representing exactly 6,737 unique physical facilities. The pipeline indexed facilities strictly by unique FACILITYID, retaining the primary feature with verified coordinates.

C. Collision-Free Entity Matching

To integrate live Federal BOP directory records with the HIFLD baseline without corrupting local municipal entities, a strict entity resolution algorithm was enforced:

- Federal BOP codes (e.g. BOP-MAR, BOP-THA) and exact institution titles were matched directly.
- Substring matching was strictly restricted to facilities classified as Federal or containing federal institution markers (USP, FCI, FDC, MDC, FMC).
- This prevented false collisions where short federal institution names would otherwise overwrite local county jails in the same municipality (e.g. USP Marion vs Marion County Jail in Illinois).

D. Data Sanitization & Formatting Rules

- Sentinel Values: Purged legacy placeholder strings ('-999', '9999', 'NOT AVAILABLE', '-1--1') into clean null values.
- Title Casing & Acronym Preservation: Standardized names to Title Case while whitelisting uppercase federal acronyms (USP, FCI, ADX, FDC, MDC, FMC, BOP, DOC, USMS, ICE).
- Postal ZIP & FIPS Formatting: Preserved leading zeroes for East Coast states and territories (e.g. '01862' for MA, '00921' for PR). Corrected upstream Pickens County Alabama FIPS typo (10107 -> 01107).
- Discrete Integer Representation: Capacities and populations are serialized as clean nullable integers without float decimals.

4. Geographic Distribution (Top 10 States)

Rank	State / Territory	Facility Count	Reported Capacity	State DOC	County Jails	Federal
1	Texas (TX)	555	311,736	136	361	36
2	Florida (FL)	417	178,382	286	103	14
3	California (CA)	413	215,306	101	267	26
4	Georgia (GA)	324	116,652	88	199	12
5	Ohio (OH)	237	76,408	40	154	7
6	New York (NY)	232	89,792	68	99	12
7	North Carolina (NC)	227	58,671	67	131	9

8	Missouri (MO)	199	50,460	31	138	7
9	Virginia (VA)	192	57,230	48	101	13
10	Illinois (IL)	187	68,284	45	109	8

5. Schema & Field Reference

Column Name	Type	Example	Description
facility_id	String	10002798	Primary unique alphanumeric facility identifier (HIFLD ID / BOP Code).
facility_name	String	Midland Co Central Det	Official standardized title-cased facility name.
jurisdiction	String	County / Local	Authority level: Federal, State, County/Local, Municipal, Private.
facility_type	String	County / Local Jail	Operational classification (State/Federal Prison, Jail, Juvenile, etc.).
security_level	String	Maximum	Security level (Maximum, Close, Medium, Minimum, Juvenile, Admin).
operational_status	String	Open	Operational status: Open, Closed, Not Available.
street_address	String	400 S Main St	Physical street address of the facility.
city	String	Midland	City where the facility is located.
state	String	TX	Two-letter US postal state/territory abbreviation (55 total).
zip_code	String	79701	5-digit or 9-digit postal ZIP code (leading zeroes preserved).
county	String	Midland	County or parish name.
county_fips	String	48329	5-digit Federal Information Processing

			Standard county code.
phone_number	String	(432) 688-4745	Standardized 10-digit telephone contact number.
website	String	https://...	Official facility or department portal URL.
latitude	Float	31.993959	WGS84 Decimal Degrees Latitude (North).
longitude	Float	-102.075419	WGS84 Decimal Degrees Longitude (West).
design_capacity	Integer	498	Official rated or design bed capacity.
population	Integer	438	Reported inmate population count.
gender	String	Male / Female	Inmate gender housing designation.
data_source	String	DHS HIFLD	Origin agency of baseline record.